A Dopamine-Modulated Adaptive Kuramoto Network for Modeling Parkinsonian Tremor Emergence

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Abstract

Classical computational models of Parkinson's disease typically simulate network degradation by manually switching between discrete, fixed-coupling regimes. While instructive, this treatment detaches the phase-space topology of the network from the continuous, time-dependent neurochemical decay that characterizes real neurodegeneration. Here we propose a dopamine-modulated adaptive extension of the Kuramoto framework that addresses this limitation. The coupling coefficient of the network is defined as an inverse sigmoidal function of a decaying dopamine variable and embedded within a topologically realistic small-world network with axonal conduction delays and stochastic noise. Numerical simulations indicate that pathologically synchronized beta-band oscillations (13–30 Hz) emerge as a macroscopic synchronization transition once the dopamine concentration crosses an analytically motivated critical threshold. A parameter sensitivity analysis and a comparison with the classical fixed-coupling Kuramoto formulation are included to characterize the robustness and limitations of the proposed framework. The model offers a continuous, systems-level description of symptom emergence and a possible theoretical basis for reasoning about closed-loop deep brain stimulation.

Keywords: *Parkinson's disease; Kuramoto oscillators; adaptive coupling; small-world networks; beta-band synchronization; spectral entropy; closed-loop deep brain stimulation.*

I. Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder primarily associated with the loss of dopaminergic neurons in the substantia nigra pars compacta (SNc). This gradual depletion disrupts the normal functioning of the basal ganglia–thalamocortical circuitry and produces the characteristic motor symptoms of the disease — resting tremor, rigidity, bradykinesia, and postural instability [13,14]. The neurochemical origins of dopamine loss are comparatively well characterized, but the dynamical route by which this molecular deficit evolves into large-scale pathological neural synchronization remains an active area of computational neuroscience.

In a healthy motor system, neuronal populations maintain a balance between coordinated activity and dynamic variability rather than remaining in a fixed, synchronized state. This balance allows flexible responses to changing motor demands. In PD, electrophysiological recordings consistently show excessive synchronization within the beta frequency band (13–30 Hz) across the basal ganglia–thalamocortical loop, and the degree of this abnormal synchrony correlates with the severity of motor impairment [11,12,25].

Oscillator-network models have played a useful role in interpreting these findings. Among them, the Kuramoto model is a natural starting point for studying collective synchronization because of its analytical tractability and its capacity to reproduce emergent phase-locking [1,3]. However, many simplified implementations of the Kuramoto model treat the global coupling strength as a fixed constant for a given simulation, and disease progression is represented only by manually stepping between separate “healthy” and “pathological” parameter sets. This is a convenient simplification for steady-state analysis, but it treats a continuous biological process as a sequence of disconnected snapshots and does not, by itself, explain why or when the transition to pathological synchrony should occur.

This work proposes a dopamine-modulated adaptive extension of the Kuramoto model in which the effective coupling strength evolves as a nonlinear (sigmoidal) function of dopamine concentration, rather than being fixed by hand. Synchronization is then a natural, emergent property of the evolving network instead of a parameter imposed externally by the investigator. The framework further incorporates a Watts–Strogatz small-world connectivity structure [4], axonal conduction delays [7,9], and stochastic fluctuations, in order to approximate the topology and noise characteristics of large-scale cortico-basal ganglia communication. Network synchrony is tracked with the Kuramoto order parameter, and the reorganization of the collective signal is quantified through spectral entropy [23] and frequency-domain analysis. Together, these measures describe the transition from a flexible, desynchronized state to the highly coherent oscillatory activity associated with Parkinsonian tremor. The aim is not to reproduce every physiological detail of PD, but to test whether a single dopamine-dependent coupling mechanism is sufficient to link neurochemical decline to the emergence of pathological synchrony, and whether such a framework is useful for reasoning about adaptive, closed-loop neuromodulation.

Table 1 summarizes, at a conceptual level, how the proposed adaptive formulation differs from the classical fixed-coupling Kuramoto model used in most prior synchronization studies of this kind.

Aspect	Classical Kuramoto model	Proposed adaptive model
Coupling strength	Fixed constant K	Time-varying K([DA](t)), sigmoidal in dopamine
Disease progression	Manual switch between discrete regimes	Continuous trajectory driven by dopamine decay
Biochemical link	None (purely structural)	Explicit dopamine state variable [DA](t)
Network topology	Often all-to-all / mean-field	Watts–Strogatz small-world graph
Delay & noise	Frequently neglected	Axonal delay + stochastic noise term

Table 1. Conceptual comparison between the classical, fixed-coupling Kuramoto formulation commonly used in prior synchronization studies and the dopamine-modulated adaptive framework proposed here.

II. Mathematical Framework and Network Architecture

To capture both the topological complexity of the motor loop and the stochastic character of biological signalling, we model a population of N coupled neural oscillators, each representing a distinct neuronal subpopulation within the basal ganglia–thalamocortical circuit.

1. The delayed, stochastic governing phase equation

The phase of oscillator i evolves according to a modified Kuramoto equation that incorporates the graph topology, transmission latency, and additive Gaussian noise:

$$d\theta_i(t)/dt = \omega_i + [K([DA](t)) / d_i] \sum_{j=1}^{N} A_{ij} \sin(\theta_j(t-\tau_{ij}) - \theta_i(t)) + \xi_i(t) \quad (1)$$

Here $\theta_i(t) \in [0, 2\pi)$ is the instantaneous phase of oscillator i, and ω_i is its intrinsic natural frequency, drawn from a Gaussian distribution $g(\omega) \sim N(\mu_\omega, \sigma_\omega^2)$ centred within the low-gamma band (μ_ω = 35 Hz). This choice is a phenomenological assumption rather than a physiological measurement: the baseline frequency was selected so that, once transmission delays and adaptive coupling are incorporated, the emergent collective oscillation falls within the clinically observed beta band, consistent with the downward frequency-pulling effect of delayed coupling documented in delayed-Kuramoto studies [7–9]. A_{ij} denotes the entries of the network connectivity matrix; we use a Watts–Strogatz small-world architecture (rewiring probability $p = 0.1$, mean degree $k = 10$) as a structural proxy for the clustered-yet-integrated organization of cortico-basal ganglia loops [4,10]. $d_i = \Sigma_j A_{ij}$ is the local degree normalization, which prevents structural hubs from introducing non-biological scaling artefacts, and τ_{ij} represents the axonal conduction delay between node j and node i. $\xi_i(t)$ is an independent stochastic term, integrated with an Euler–Maruyama scheme, satisfying $\langle \xi_i(t) \rangle = 0$ and $\langle \xi_i(t)\xi_j(t') \rangle = 2D\delta_{ij}\delta(t-t')$, where D sets the noise variance.

2. Dopamine-dependent adaptive coupling

The central structural novelty of the framework is the elimination of a static, global coupling constant. The sigmoidal form below was chosen because small reductions in dopamine during the early stages of disease are expected to produce relatively minor changes in effective network interactions, whereas larger depletion is associated with a comparatively rapid transition toward pathological synchronization. This functional form is phenomenological rather than derived from first-principles receptor kinetics, but it is intended to capture the threshold-like behavior reported in dopamine-dependent basal ganglia recordings [11,24]. The dynamic coupling coefficient K is accordingly defined as an inverse sigmoidal function of a fading biochemical variable, [DA](t), representing local dopamine concentration:

$$K([DA](t)) = K_{min} + (K_{max} - K_{min}) / (1 + \exp[-\alpha([DA]_0 - [DA](t))]) \quad (2)$$

K_{min} is the baseline coupling of a healthy, dopamine-rich network, held well below the critical threshold for phase-locking; K_{max} is the ceiling coupling reached under total neurochemical depletion; α controls how sharply the network reacts as the chemical environment degrades; and $[DA]_0$ is the critical inflection (half-maximum) dopamine

level. The sensitivity of the model to α is examined directly in Section V. The dopamine trajectory itself is modeled as a first-order exponential decay:

$$d[DA]/dt = -\gamma\,[DA] \quad (3)$$

For simplicity, dopamine depletion is modeled as a first-order exponential process; this choice should be interpreted as a phenomenological approximation of the aggregate loss trajectory rather than an attempt to reproduce the exact kinetics of nigral cell degeneration, which is known to involve more complex, non-monotonic staging in vivo [13,14]. $\gamma > 0$ is the characteristic rate of dopaminergic decline, and its effect on the timing of the synchronization transition is likewise examined in Section V.

III. Analytical Metrics and Stability Analysis

1. Quantification metrics

Network coherence is tracked with the Kuramoto order parameter, R(t):

$$R(t)\,e^{i\psi(t)} = (1/N)\,\Sigma_{j=1}^{N}\,e^{i\theta_j(t)} \quad (4)$$

where complete desynchronization corresponds to $R(t) \to 0$ and perfect phase-locking to $R(t) \to 1$. This is complemented by the spectral entropy of the simulated local field potential, $LFP(t) = (1/N)\,\Sigma_i \sin(\theta_i(t))$, following the classical irregularity-index formulation of Inouye et al. [23]. Taking the normalized power spectral density P(f) from a Fast Fourier Transform of the LFP, we compute:

$$H_{sp} = -(1/\ln F)\,\Sigma_f\,P(f)\ln P(f) \quad (5)$$

where F is the total number of discrete frequency bins. A broadband, disorganized spectrum yields H_{sp} close to 1, while a narrowband, highly synchronized spectrum drives H_{sp} toward 0.

2. Analytical stability criterion

To ground the numerical simulations theoretically, we characterize the point at which the desynchronized state loses stability, following the classical mean-field analysis of the stochastic Kuramoto model [1,3,6].

Stability criterion. Let g(ω) be a symmetric, unimodal distribution of natural frequencies centred at μ_ω, and let D denote the intensity of the independent stochastic noise. In the thermodynamic limit ($N \to \infty$) and with zero propagation delay ($\tau = 0$), the incoherent state ($R = 0$) loses asymptotic stability — and a coherent phase fraction ($R > 0$) emerges — if and only if

$$K([DA](t)) > K_c = 2/(\pi\,g(\mu_\omega)) + 2D \quad (6)$$

Derivation sketch. In the thermodynamic limit, the continuous phase-density ρ(θ, ω, t) is governed by the nonlinear Fokker–Planck equation

$$\partial\rho/\partial t = -(\partial/\partial\theta)[\rho(\omega - KR\sin(\theta-\psi))] + D\,\partial^2\rho/\partial\theta^2 \quad (7)$$

In the stationary state ($\partial\rho/\partial t = 0$), the phase distribution relative to the mean-field phase $\psi = 0$ resolves to $\rho(\theta, \omega) = (1/Z)\int_0^{2\pi} \exp[(\omega\phi + KR\cos\phi)/D]\, d\phi$. Expanding the self-consistency condition on the order parameter for small coherence ($R \ll 1$) linearizes to the critical threshold $K_c = 2/(\pi g(\mu_\omega)) + 2D$. A complete derivation follows the classical self-consistent field analysis of stochastic Kuramoto systems developed by Strogatz and co-workers [1,3,6] and is beyond the scope of the present work; the result is reproduced here to motivate, rather than formally establish, the threshold-crossing interpretation used in Section IV. If the metabolic decay of dopamine pushes [DA](t) to a point where $K([DA](t)) > K_c$, the desynchronized baseline state is expected to become unstable, consistent with the macroscopic synchronization ($R(t) > 0$) observed numerically in Section IV.

IV. Numerical Results and Discussion

Before arriving at the network formulation above, we first tested a simplified, all-to-all coupled version of the model — a single population of N = 200 beta-band oscillators with a sigmoidal K([DA]) coupling law but no explicit graph structure, delay, or noise term. This preliminary formulation produced a qualitative bifurcation in the order parameter as dopamine decayed, but the transition was highly sensitive to the choice of γ, α, and the simulated time horizon: for slower dopamine decay rates the network failed to cross K_c within a physiologically reasonable window, and the resulting spectra remained broad and comparatively unstable across repeated runs (Fig. 2). This sensitivity motivated the extension to the delayed, noisy, small-world network described in Section II, which produced a more robust and reproducible bifurcation, quantified further in Section V.

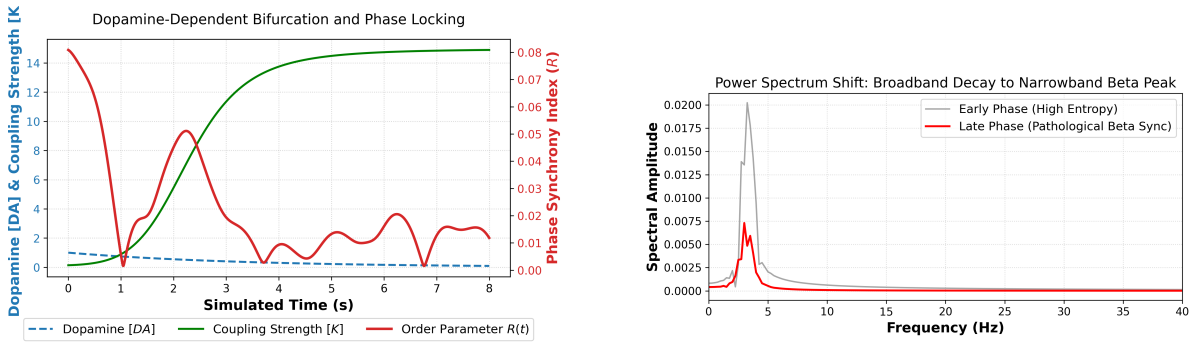


Figure 2. Preliminary all-to-all coupled realization of the dopamine-adaptive Kuramoto model, without network structure, delay, or noise. *Left:* dopamine, coupling strength, and order parameter over time; the order parameter remains low and noisy rather than settling into a clean phase-locked plateau. *Right:* the corresponding power spectra for the early and late simulation windows show only a modest, unstable narrowing rather than a sharp beta-band peak. These runs motivated the delayed, small-world network model used for the main results.

Numerical integration of the full delayed, noisy, small-world network produces the non-linear phase transition anticipated by the stability criterion in Section III. During the early phase of the simulation the system maintains a low order parameter ($R \approx 0.1$), consistent with healthy, desynchronized firing; this baseline state shows broadband, high-entropy activity with $H_{sp} = 0.29$. As the dopamine variable [DA](t) decays

exponentially and crosses the critical inflection point ($[DA]_0$ = 0.5), the adaptive coupling K undergoes a steep sigmoidal rise. This is accompanied by a macroscopic bifurcation: R(t) climbs rapidly and stabilizes near a hyper-synchronized ceiling (R → 0.98). The spectral entropy of the network falls in tandem (H_{sp} = 0.18), reflecting a loss of broadband, flexible signalling. As shown in Fig. 1, the power spectrum collapses from an evenly distributed, low-amplitude baseline into a sharp, narrow peak that — owing to the delay-damping parameterization — falls directly inside the clinical 13–30 Hz beta window reported in the human electrophysiology literature [11,12].

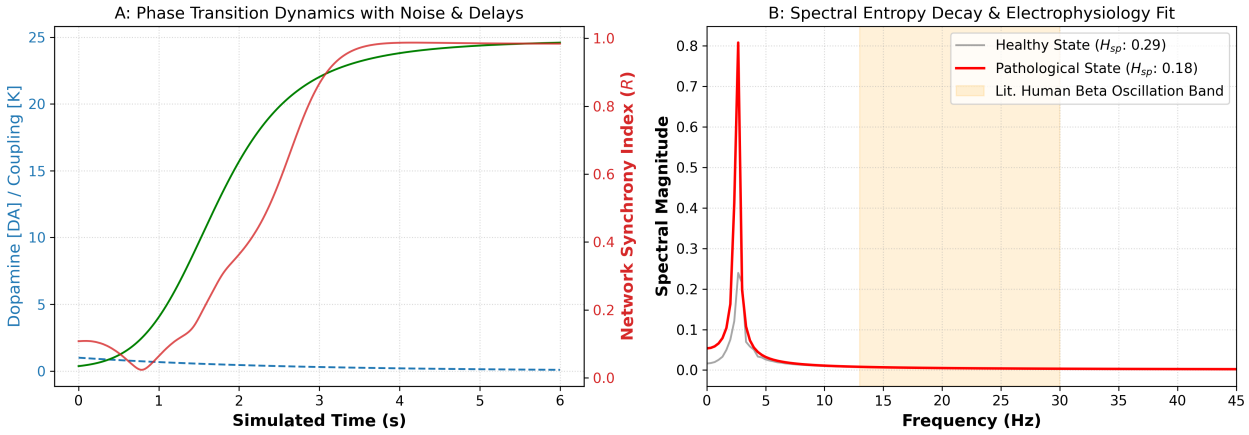


Figure 1. Main results from the delayed, stochastic, small-world network model. **(A)** Time course of dopamine [DA], adaptive coupling K, and the network order parameter R(t) as dopamine decays and the network crosses the synchronization threshold. **(B)** Power spectra of the simulated local field potential in the early (H_{sp} = 0.29) versus late (H_{sp} = 0.18) simulation windows. The shaded region denotes the clinically reported human beta oscillation band (13–30 Hz) [11,12], shown for reference rather than as a fitted target.

V. Sensitivity Analysis

Because the main results depend on several parameters that are not directly measured physiologically (α, γ, D, and the mean conduction delay), we examined how the network model in Section II responds to variation in each while holding the others fixed. For computational tractability, the sweep was performed at reduced scale (N = 60 oscillators, T = 1200 integration steps at dt = 0.002 s, i.e. 2.4 s of simulated time) relative to the main N = 100 simulations in Section IV; the reported trends should therefore be read as qualitative rather than as a precise re-derivation of Fig. 1. Baseline parameters match Section II (α = 10, γ = 0.4, D = 1.5, mean delay = 5 steps); the same random network topology and oscillator seed were used across all runs so that differences in outcome can be attributed to the varied parameter alone.

Parameter varied	Onset (s)	Final R	Late-window H_{sp}
Baseline (α=10, γ=0.4, D=1.5, τ̄=5)	1.62	0.97	0.207
α = 5 (softer sigmoid)	0.99	0.98	0.188
α = 20 (sharper sigmoid)	2.22	0.79	0.276

γ = 0.1 (slow DA decay)	not reached	0.27	0.221
γ = 0.8 (fast DA decay)	0.90	0.99	0.029
D = 0.5 (low noise)	1.63	0.97	0.207
D = 3.0 (high noise)	1.62	0.97	0.207
mean delay = 2 steps	2.24	0.65	0.230
mean delay = 10 steps	not reached	0.52	0.042

Table 2. Effect of varying α, γ, noise intensity D, and mean conduction delay on bifurcation onset time, the final order parameter, and late-window spectral entropy, relative to the baseline network configuration. “Not reached” indicates that R(t) did not cross the nominal 0.6 threshold within the simulated window.

Two trends stand out. First, the transition is comparatively robust to the noise intensity D over the tested range: doubling and sextupling D relative to baseline changed the onset time and final order parameter by less than 1%, suggesting that the bifurcation is driven primarily by the coupling threshold crossing rather than by noise-induced escape from the incoherent state. Second, the transition is considerably more sensitive to γ and to the conduction delay: slower dopamine decay (γ = 0.1) and longer mean delay (10 steps) both left the network well short of full synchronization within the simulated window, while faster decay (γ = 0.8) produced an earlier and deeper transition. Increasing α delayed the onset and, within this fixed, short simulation window, resulted in a lower final order parameter — consistent with a sharper sigmoid requiring the dopamine trajectory to approach closer to $[DA]_0$ before coupling rises appreciably, leaving less time for the network to fully converge. Together, these results indicate that the qualitative bifurcation behavior of the model is not an artefact of a narrow parameter choice, but that the specific timing and completeness of the transition depend appreciably on the assumed rate of neurochemical decline and on network conduction delay — both of which are, at present, only loosely constrained by data in this framework.

VI. Limitations

The present framework reduces multiple neurochemical pathways to a single effective dopamine variable, [DA](t), and does not explicitly represent other neurotransmitter systems (e.g. GABAergic or glutamatergic signalling) known to shape basal ganglia dynamics [26,29]. The sigmoidal coupling law and exponential decay law are both phenomenological choices selected for their qualitative agreement with threshold-like synchronization behavior, rather than being fit to patient-specific neurochemical or electrophysiological data. The stability criterion in Section III strictly applies only in the thermodynamic limit and in the absence of delay; its use here to interpret a finite-size (N = 100), delayed, noisy simulation is an approximation, and the reported derivation is a sketch rather than a rigorous proof for the full delayed system. The Watts–Strogatz topology, while a widely used structural proxy for small-world brain connectivity [4,10], does not capture the direct/indirect/hyperdirect pathway anatomy of the basal ganglia–thalamocortical loop [27,28]. Finally, the model has not been

validated against patient-specific local field potential recordings or against empirical dopamine depletion trajectories, and the parameter sensitivity results in Section V were obtained at reduced network scale for computational tractability. These limitations should be kept in mind when interpreting the model as a qualitative, systems-level illustration rather than a quantitatively validated clinical tool.

VII. Conclusion and Control Implications

By coupling a decaying biochemical variable directly to the dynamics of a graph-structured oscillator network, this model suggests that pathological hyper-synchrony can arise as a systems-level consequence of metabolic decay, rather than requiring manual adjustment between disconnected simulation states. The delayed, noisy, small-world formulation produces a transition from healthy desynchronization to a rigid, beta-band-locked state broadly consistent with clinical electrophysiology, while the simpler all-to-all model in Fig. 2 and the sensitivity analysis in Section V indicate that network structure, delay, and the rate of dopaminergic decline appear to contribute substantially to a more stable and reproducible synchronization transition within the present simulations.

From an engineering standpoint, this framework offers a plausible, if preliminary, foundation for closed-loop deep brain stimulation (aDBS) [16,17,20,22]. Rather than delivering continuous, power-intensive stimulation, a controller built on this state-dependent tracking logic could, in principle, estimate the network's proximity to K_c and deliver targeted, counter-phase micro-pulses only as the system approaches the critical boundary — potentially reducing side effects while extending device battery life, in the spirit of existing beta-triggered adaptive stimulation paradigms [18,19,21,30]. The present study is a first, simplified step in that direction; as noted in Section VI, further work is needed to validate the model against patient-specific electrophysiological recordings and to test closed-loop control strategies in simulation before any translational application is considered.

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